

Institute of Actuaries of India

Subject CT5 – General Insurance, Life and Health Contingencies

March 2018 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) The first method is based on the assumption that, for integer x and $0 \leq t \leq 1$, the function ${}_t p_x \mu_{x+t}$ is a constant. The second method of approximation is based on the assumption of a constant force of mortality. That is, for integer x and $0 \leq t \leq 1$, $\mu_{x+t} = \mu = \text{constant}$

[2]

ii)

a) ${}_4 p_{65.5} = {}_{0.5} p_{65.5} \times {}_3 p_{66} \times {}_{0.5} p_{69}$

$${}_3 p_{66} = \frac{l_{69}}{l_{66}} = \frac{9473.745}{9658.285} = 0.980893$$

Under UDD,

$${}_{0.5} p_{65.5} = \frac{p_{65}}{{}_{0.5} p_{65}} = \frac{1 - q_{65}}{1 - {}_{0.5} q_{65}} = \frac{1 - 0.004681}{1 - 0.5 \times 0.004681} = 0.997654$$

$${}_{0.5} p_{69} = 1 - {}_{0.5} q_{69} = 1 - 0.5 (0.008563) = 0.995719$$

$$\text{Therefore, } {}_4 p_{65.5} = 0.997654 \times 0.980893 \times 0.995719 = 0.9744021$$

[2]

b) Under CFM,

$${}_{0.5} p_{65.5} = (p_{65})^{0.5} = (1 - 0.004681)^{0.5} = 0.997657$$

$${}_{0.5} p_{69} = (p_{69})^{0.5} = (1 - 0.008563)^{0.5} = 0.995709$$

$$\text{Therefore, } {}_4 p_{65.5} = 0.997657 \times 0.980893 \times 0.995709 = 0.974396$$

[2]

[6 Marks]**Solution 2:**

- i) Different categories of direct expenses are:-

- Initial expenses – those arising when policies are issued – e.g. commission or acquisition expenses
- Renewal expenses – those arising regularly during the policy term – e.g. administrative expenses
- Termination expenses – those arising when the policy terminates as a result of an insured contingency – e.g. claim expenses

[3]

- ii) The main categories of costs affected by inflation are:

- salaries and salary-related expenses,
- buildings and other property costs,
- computing and associated costs, and

- costs associated with the investment of funds.

Expenses being valued are adjusted for expected inflation in order to allow in the premium. Publicly available data e.g. retail price index, national average earnings index, and similar data internal to the insurance company can be used to forecast the expected general price, future wage or specific price inflation.

[4]

[7 Marks]

Solution 3:

- i) The expected present value for a standard life policy is:

$$\begin{aligned}
 5,00,000 \times \bar{A}_{40:10}^{-1} &= \frac{i}{\delta} A_{40:10}^1 = \frac{i}{\delta} (A_{40} - v^{10} {}_{10}p_{40} A_{50}) = \frac{i}{\delta} \left(A_{40} - v^{10} \frac{l_{50}}{l_{40}} A_{50} \right) \\
 &= 5,00,000 \times \frac{0.06}{\ln 1.06} \left(0.12313 - 0.55839 \times \frac{9712.0728}{9856.2863} \times 0.20508 \right) \\
 &= 5,00,000 \times 0.010596 = 5,298
 \end{aligned}$$

[4]

- ii) Since the mortality for impaired life is 5 times the mortality of standard life, he is eligible for a coverage of 1,00,000 against 5,00,000 for a standard life and a policy term of 2 years against 10 years for a standard life.

The expected present value for an impaired life policy is:

EPV equation based on first principle is,

$1,00,000(vq_{40}^* + v^2p_{40}^*q_{41}^*)$ where * denotes impaired mortality

$$\begin{aligned}
 p_{40}^* &= \exp\left(-\int_0^1 \mu_{40+t}^* dt\right) = \exp\left(-\int_0^1 5\mu_{40+t} dt\right) = (p_{40})^5 \\
 &= 1,00,000\left(v \times [1 - (p_{40})^5] + v^2 (p_{40})^5 [1 - (p_{41})^5]\right) \\
 &= 1,00,000\left(0.94340 \times [1 - (0.999063)^5] + 0.89 \times (0.999063)^5 [1 - (0.998986)^5]\right) \\
 &= 889.36
 \end{aligned}$$

[3]

[7 Marks]

Solution 4:

- i) Assume a policy issued t years ago to a life then aged x, with sum assured of S payable at the end of the year of death. Also assuming, no survival benefit is due if the life survives to t + 1

Let ${}_tV$ be the reserve at time t. Then we define the death strain in the policy year t to t + 1 to be the random variable, DS, say,

$$DS = 0, \text{ if the life survives to } t+1$$

$$= S - {}_{t+1}V, \text{ if the life dies in the year } t \text{ to } t + 1$$

The maximum death strain, $(S - {}_{t+1}V)$ is called the death strain at risk. [2]

ii)

a) Premium for endowment type of policy:

- Annual premium for endowment assurance with 500,000 sum assured given by:

$$P = \frac{5,00,000}{\ddot{a}_{40:10}} \times A_{40:20}$$

$$\text{Where, } \ddot{a}_{40:10} = 1 - v^{10} \left(\frac{l_{50}}{l_{40}} \right) + (\ddot{a}_{40} - 1) - v^{10} \left(\frac{l_{50}}{l_{40}} \right) \times (\ddot{a}_{50} - 1)$$

$$= 1 - 0.67556 \times \left(\frac{9712.0728}{9856.2863} \right) + (20.005 - 1) - 0.67556 \times \left(\frac{9712.0728}{9856.2863} \right) \times (17.444 - 1)$$

$$= 8.3930$$

$$P = \frac{5,00,000}{8.3930} \times 0.46433 = 27,662$$

- Reserves at the end of 6th year for endowment assurance:

$${}_6V = 5,00,000 \times A_{46:14} - P \ddot{a}_{46:4}$$

$$\text{Where, } \ddot{a}_{46:4} = 3.7660$$

$${}_6V = 5,00,000 \times 0.58393 - 27,662 \times 3.7660 = 1,87,790$$

- Reserves at the end of 11th year for term assurance:

$${}_{11}V = 1,000,000 \times A^1_{41:19}$$

$$\text{Where, } A^1_{41:19} = 0.03476$$

$${}_{11}V = 1,000,000 \times 0.03476 = 34,760$$

- Sum at risk for each type of policy:

During 2005 - Endowment assurance = DSAR = 5,00,000 - 1,87,790 = 3,122,10

During 2010 - Term assurance = DSAR = 1,000,000 - 34,760 = 9,65,240

[9]

b) Mortality profit = Expected death strain (EDS) – Actual death strain (ADS)

For endowment assurance,

$$\text{EDS} = 19770 * q_{45} * 3,122,10 = 9,042,556$$

$$ADS = 35 * 3,122,10 = 10,927,353$$

$$\text{Mortality profit} = -1,884,797 \text{ (it's a loss)}$$

For term assurance,

$$EDS = 9775 * q_{40} * 9,65,240 = 8,840,802$$

$$ADS = 3 * 9,65,240 = 2,895,720$$

$$\text{Mortality profit} = 5,945,082$$

Hence, mortality loss for endowment policy is -1,884,797 and mortality profit for term is 5,945,082

[3]

c) Though mortality experience for term assurance is better even after taking into account delay in claims reporting, overall, company is experiencing a mortality loss. For endowment assurance, actual deaths are higher by 21%. Further investigation required to understand poor mortality experience for endowment assurance. Also company needs to look into reasons for delay in claims observed for term assurance along with ensuring sufficient reserve is held for late claims.

[1]

[15 Marks]

Solution 5:

i) Gross premium for the compound reversionary policy:

$$\text{EPV premiums} = P \ddot{a}_{[40]:25} = 15.887P$$

$$\text{EPV expenses} = 1200 + 500 * \bar{A}_{[40]} + 2\% * P * (\ddot{a}_{[40]:25} - 1)$$

$$= 1200 + 500 * (1.04)^{0.5} * 0.23041 + 0.02 * P * (15.887 - 1)$$

$$= 1317.487 + 0.29774P$$

$$\text{EPV benefits} = 2,00,000(v^{0.5}q_{[40]} + v^{1.5}(1.04)_1q_{[40]} + v^{2.5}(1.04)^2_2q_{[40]} + \dots)$$

$$= \frac{200000}{1.04^{0.5}}(q_{[40]} + (1.04)_1q_{[40]} + (1.04)^2_2q_{[40]} + \dots)$$

$$= \frac{200000}{1.04^{0.5}} = 1,96,116.14$$

Using the equivalence principle,

$$15.887P = 1,96,116.14 + 1317.487 + 0.29774P$$

$$P = 12,664$$

[4]

ii) Revised bonus rates for simple reversionary policy:

Since company plans to charge the same premium rate, simple reversionary bonus rate can be calculated by equating the EPV of benefits of the two policy. Let's consider $x\%$ as simple bonus rate:

$$\text{EPV of benefits} = 2,00,000 \times x\% \left(I \bar{A} \right)_{[40]} + 2,00,000 \times (1 - x\%) \bar{A}_{[40]}$$

$$= (1.04^{0.5}) [2,00,000 \times x\% (IA)_{[40]} + 2,00,000 \times (1 - x\%) A_{[40]}]$$

EPV of benefits to be equated with the EPV of benefits as per compound reversionary bonus

$$\text{Therefore, } (1.04^{0.5}) [2,00,000 \times x\% (IA)_{[40]} + 2,00,000 \times (1 - x\%) A_{[40]}] = 1,96,116.14$$

Solving equation, $X\% = 9\%$ is the simple reversionary rate.

[4]

iii) Premium as per net premium basis:

$$P \ddot{a}_{40:25} = 2,00,000 \bar{A}_{40}$$

$$P = 2,960.54$$

Net premium reserve at time 10,

$$_{10}V = 3,00,000 \bar{A}_{50} - P \ddot{a}_{50:15}$$

$$= 3,00,000 * 1.04^{0.5} * 0.32907 - 2,960.54 * 11.253$$

$$= 67,361.10$$

[4]

[12 Marks]

Solution 6:

Let force of decrement due to death be represented as μ_x , force of decrement due to diagnosis of cancer as σ_x .

Value of the benefits on death and Cancer is

$$V = 100,000 \int_0^{20} v^t {}_t p_{45} {}^{hh}(\mu_{45+t} + \sigma_{45+t}) dt$$

$$100,000 \int_0^{20} e^{-\ln(1.05)t} {}_t p_{45+t} {}^{hh}0.008 dt$$

$${}_tp^{hh} = \exp\left(-\int_{45}^{45+t} (\mu_s + \sigma_s) ds\right)$$

$$= \exp\left(-\int_{45}^{45+t} 0.008 ds\right)$$

$$= e^{-0.008t}$$

The question refers to decrements due to death and does not explicitly state whether deaths after cancer are included in this or not. Hence marks would be given to both kinds of interpretation i.e 0.006 taken as death excluding cancer and 0.006 including death after cancer which means death excluding after cancer is $0.006 - 0.002 * 0.01 = 0.00598$

$$\text{Therefore, } V = 800 * \int_0^{20} e^{-\ln(1.05)t} \cdot e^{-0.008t} dt$$

$$800 * \int_0^{20} e^{-\ln(1.05)t} \cdot e^{-0.008t} dt$$

$$= 800 \int_0^{20} e^{-0.05679t} dt$$

$$= 800 * [-e^{-0.05679t} / 0.05679]_0^{20}$$

$$= 800 * (-5.65531 + 17.60873) = 9562.8$$

Value of the Return of Premium

$$= P * e^{\int_{45}^{65} (0.006 + 0.002) dx} * v^{20}$$

$$= P * e^{-0.008 * (65 - 45)} * v^{20}$$

$$= P * e^{-0.16} * 0.37689$$

Premium = Value of Benefits on Death and Cancer + Value of Return of Premium + Commission + Expenses + Profit Margin

$$\text{So the Premium} = P = 9562.8 + 0.321164 * P + 0.02 * P + 0.02 * P + 0.1 * P$$

$$\text{Premium} = 17,747.15$$

[8 Marks]

Solution 7:**i) Unit fund projection**

	Premium	Allocated Premium	Fund after allocn	Fund before charges	Fund after policy admn charge	Fund after fund mgmt charge	Fund at year end
Yr 1	50,000	45,000	45,000	47,700	47,100	46,629	46,629
Yr 2	50,000	47,500	94,129	99,777	99,177	98,185	98,185
Yr 3	50,000	47,500	1,45,685	1,54,426	1,53,826	1,52,288	1,52,288

The probabilities are as estimated below

	Indep prob of death	Indep prob of surrender	Prob of in-force	Prob of in force per policy issued
Year				
1	0.000620	0.2	0.799380	0.799380
2	0.000652	0.1	0.899348	0.718921
3	0.000689	0.1	0.899312	0.646534

The non-unit cashflows are given below

	Prem Alloc chg	Expenses	Commission	Interest	Policy admin charge	Fund mgmt charge	Cost of death benefit	Cost of addl units	Surr charge	Profit CF
Year	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	5000	(5,000)	(7,500)	-450	600	471	-64		933	-6011
2	2500		(1,000)	90	600	992	-34			3148
3	2500		(1,000)	90	600	1,538	0	-7614		-3886

Year t	Profit in Year t	Profit Signature	Disc Profits
1	(6,011)	(6,099)	(5,565)

2	3,148	2,516	2,157
3	(3,886)	(2,794)	(2,218)

The net present value of profits is -5,626.

The profit margin is -11%

[9]

- ii) From the non-unit cashflows it is seen that there is a huge strain at the end of the 3rd policy year as there is not enough positive cashflows to fund the addition of units.

From a sound financial performance perspective the company should hold reserves such that there is no strain in the future. The company needs to set up non unit reserves such that these reserves can meet the funding strain.

Zeroised non unit reserves based on the amount required at the end of every policy year should be held.

[2]

- iii) The profit margin is a negative 11% compared to the expected 15%. Hence the charging structure needs to be changed and profit tested again till the desired profit criterion is met.

The changes that can be made are

- Increase the premium allocation charge by 5% or so every year
- Remove the benefit of adding 5% of the fund value at the end of 3rd policy anniversary
- Increase the policy admin charge or fund management charge (from 1% to 2%)
- Reduce the commission payable in 1st year to 5% and remove renewal commissions

[3]

[14 Marks]

Solution 8:

The reserve at end of Year 3 on the reserving basis to be calculated on a recursive basis.

The amount required at end of Year 5 is the amount for Maturity Benefit and Death Benefit

Maturity amount = Sum Assured plus bonus @ 5% of Sum assured per year = 45000 + 45000*5%*5

=56,250

Death benefit = Sum Assured plus bonuses = 56,250

Reserve = ${}_5V = 56,250$

Reserve required at end of 4th year = ${}_4V = (56,250)/(1+4\%) - (10,000 - 100 - 500) = 44,687$

Reserve required at end of 3rd year = ${}_3V = (44,687 \times 0.998 + 108)/(1+4\%) - (10,000 - 100 - 500) = 33,586$

Reserve of 33,586 is to be maintained for the altered policy. The altered Sum assured 'X' should be derived such that

Reserve @ end of 4th year = $(X' + 6750)/(1.04) + 100 = Y$

Reserve @ end of 3rd year = $(Y \times 0.998 + (X' + 6750) \times 0.002)/(1+4\%) + 100 = 33,586$

$$(X'/1.04 + 6750/1.04 + 100) \times 0.998 + 0.002 \times X' = 34811.94$$

$$X'/1.04 \times 0.998 + 0.002 \times X' = 28234.74$$

$$X' = 29,362$$

[5 Marks]

Solution 9:

- i) Assuming that retirements happen uniformly during the age x to x+1, the age at retirement can be taken as $x+1/2$

$$PS_{x+1/2} = \frac{1}{3} * S \left[\frac{s_{x-1/2} + s_{x-1/2} + s_{x-2 1/2}}{s_{x-1}} \right]$$

where s_x is the salary scale.

$$\text{Let's define } z_x \text{ as } \frac{1}{3} * [s_{x-1} + s_{x-2} + s_{x-3}]$$

Then

$$\text{Pensionable Salary is } PS_{x+1/2} = S z_{x+1/2} / s_{x-1}$$

The total pension for $(m+1/2)$ years of service is

$$(m+1/2)/60 * S z_{x+1/2} / s_{x-1} \quad [3]$$

ii) The Past Service liability is the value of the liability related to 'm' years of service

$$PSL = m/60 * S_{z_{x+1/2}} / s_{x-1} * a_{x+\frac{1}{2}}^r * \frac{v^{x+1/2}}{v^x} * \frac{r^x}{l^x}$$

where $a_{x+\frac{1}{2}}^r$ is the value of the annuity payable for retirals, r^x and l^x are values from the pension table used to derive probability of retirement between x and x+1.

The Future Service liability relates to the future service in the year of retirement – assuming retirements happen uniformly during the year, average service is ½ year.

$$FSL = 1/2/60 * S_{z_{x+1/2}} / s_{x-1} * a_{x+\frac{1}{2}}^r * \frac{v^{x+1/2}}{v^x} * \frac{r^x}{l^x} \quad [3]$$

iii) If retirement happens at anytime after valuation date, then these are events happening between age x and x+1, x+1 and x+2 etc upto the Normal retirement age of 65

The liability is the sum of liability if retirement were to happen between age x and 65 and liability if retirement were to happen at age 65

a) Assuming that the retirements happen on an average at age $x+t+1/2$ where t ranges from 0 to $65 - x - 1$

The Pension liability for Past Service would be summation of expected liability for each year of retirement with the associated probabilities.

$$\sum_{t=0}^{65-x-1} \frac{1}{60} * S_{\frac{z_{x+t+1/2}}{s_{x-1}}} * a_{x+t+\frac{1}{2}}^r * \frac{v^{x+t+1/2}}{v^x} * \frac{r^{x+t}}{l^x}$$

b) For retirement at age 65 the Past Service liability is

$$\frac{1}{60} S_{\frac{z_{65}}{s_{x-1}}} a_{x+t+\frac{1}{2}}^r * \frac{v^{65}}{v^x} * \frac{r^{65}}{l^x}$$

The total liability is a+b

[2]

[8 Marks]

Solution 10:

i)

- Class selection: It refers to the selection specified by a certain category of a particular characteristic like gender, age, occupation. The selection is permanent with respect to its mortality/morbidity

Examples in a pension fund – mortality of members retiring at normal retirement age or early (but in good health) is expected to be lower than mortality of members retiring due to ill-health. The mortality of working members is expected to vary by age, gender and type of occupation.

- Temporary initial selection: The process of selection of lives defined by a specified event happening to all members of the group at a particular age. It is called temporary as the effect of selection wears over time

Examples in pension fund

Ill-health retirals at a certain age – Since members are retiring due to ill-health these are expected to experience higher (and hence similar) mortality/morbidity rates immediately after retirement and the effect is expected to wear off after some time

- Time selection

The process by which experience varies over time is called time selection.

In a pension fund, withdrawals due to redundancy tend to vary over time as economic conditions vary, and is an example of time selection.

[4]

- ii) Summary mortality indices are single-figure indices which facilitate the comparison of mortality experience in two groups or in the same group over time in a simpler way. All summary measures are weighted averages of the age-specific rates or some function of age-specific rates. The advantage is the use of a single figure as compared to a set of age-specific factors.

Key word – simplicity, comparison of mortality

Crude mortality rate: It's a summary measure defined as Actual deaths/Total exposed to risk. It's a weighted average of central rate of mortality with exposed to risk as the weights.

Main advantage is knowledge of mortality rates and exposed to risk for separate ages is not required.

Main disadvantage is that the differences in age structure between the groups being compared will be confounded with differences in mortality rates

The above disadvantage can be dealt with by standardising the mortality rates - this is done by applying weighting factors to each age group to compensate for the different population

structures. Populations may need to be standardised by age, sex or some other risk factor like occupation.

Directly standardised mortality rate

This summary measure is the weighted average of mortality rates using the exposed to risk of a standard population

The death rates are standardised by working out how many deaths we would have had if the proportions at each age had been the same as in a standard population.

The advantage is that it's a weighted average of mortality only and hence does not reflect differences in age structures between populations

Its main disadvantage is that it requires age specific mortality rates $m_{x,t}$ for its calculation.

Indirectly standardised rate uses an approximation wherein a area comparability factor is used to overcome the need for age-specific rates

Indirectly std mortality rate = Crude mortality rate for standard population * Expected deaths in population for standard population / Actual deaths in population

Advantage is that it does not require age-specific mortality rates and is a good enough approximation of the standardised rate. [5]

- iii) The **crude birth rate** is obtained by dividing the total number of births by the total exposed to risk, which gives

$$CBR = 495/530000 = 0.000933 \text{ or } 0.933 \text{ per thousand}$$

Standardised birth rate

The standardised birth rate is obtained by first calculating the number of births that would be expected in the whole country based on the birth rates for the State, which gives:

$$\text{Exp. Births} = 20 / 20000 * 150,000 + 150/125,000 * 1,250,000 + 200/150,000 * 1,200,000 + 75/135,000 * 1,485,000 + 50/100,000 * 900,000 = 4,525$$

This is then divided by the exposed to risk for the whole country:

$$\text{Std. Birth Rate} = 4,525/4,985,000 = 0.000908 = 0.908 \text{ per thousand}$$

Standardised birth ratio

For the standardised birth ratio, we first need to find the number of births that would be expected in State if national birth rates applied, which gives:

$$\text{Exp. births} = 120 / 150,000 * 20,000 + 1,650 / 1,250,000 * 125,000 + 2000 / 1,200,000 * 150,000 + 700 / 1,485,000 * 135,000 + 350 / 900,000 * 100,000 = 533.525$$

The standardised birth ratio is then found by dividing the actual number of birth with the expected births b

$$\text{SBR} = 495 / 533.525 = 0.92779 \quad [3]$$

- iv) The comparison of crude birth rates of State is lower than that of the Country. Hence the observation is right.

However the comparison of age-specific Crude birth rates shows that the rates in age group 30+ (those who were 20-30 10 yrs back) is higher than that in the country. Hence the attribution to the awareness programme may not be correct.

[2]

[14 Marks]**Solution 11:****i) Probability of dying between age 50 and 51**

Assuming withdrawals, age-related and ill-health related retirements and deaths happen uniformly during the year, prob is calculated as deaths/exposed

$$= 9 / (19000 - 0.5 * (4100 + 185 + 50)) = 0.000534 \quad [1]$$

ii) Probability of dying retiring between age 50 and 53

= prob of retiring between age 50 and 51 + prob of remaining in service till ag 51 * prob of retiring between age 51 and 52 and prob of remaining in service till ag 52 * prob of retiring between age 52 and 53

$$= 0.013868 + 0.019525 * 0.771368 + 0.025583 * 0.641307 = 0.045336 \quad [1.5]$$

iii) Prob of not receiving any benefit :

This would happen if a member has joined the scheme between age 50 and 51 and leaves after reaching 51, but before completing 1 year of service.

Assuming new members enter uniformly during the year and withdrawals happen uniformly during the year, a recently joined member and aged 51 would have completed 6 months of service and will get the benefit if completes another 6 months between age 51 and 52

New members joined between age 50 and 51 = 130

Prob of member aged 51 not receiving any benefit = prob of being a new member * prob of leaving service * $\frac{1}{2}$

$$= 0.01625 * 0.151420 * \frac{1}{2} = 0.001230$$

[1.5]

[4 Marks]
